Time Series Analysis

Chapter 6 – ARCH and GARCH models

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Chapter 1

ARCH and GARCH models

Introduction

The ARMA family of Chapter **??** captures the dynamics of the *conditional mean*, but assumes a constant variance. On financial returns, on macroeconomic surprises and on many physical measurements, this assumption is violated: large absolute deviations cluster in time, and the variance itself is predictable. This chapter introduces the *autoregressive conditional heteroskedasticity* (ARCH) family that models the conditional variance directly, generalises it to GARCH(p, q), derives the stationarity conditions, presents the maximum-likelihood estimator, and applies the model to volatility forecasting and Value-at-Risk.

1.1 Stylised facts of financial returns

Remark – Empirical regularities

For daily log-returns $r_t = \log(P_t/P_{t-1})$ on broad equity indices, currencies and commodities:

1. $\mathbb{E}[r_t] \approx 0$ (small drift).
2. ACF of r_t: nearly flat – returns are approximately serially uncorrelated.
3. ACF of r_t^2 (or $|r_t|$): *strongly* positive at many lags – volatility is autocorrelated.
4. Marginal kurtosis $\widehat{\kappa} > 3$: heavier tails than Gaussian.
5. Leverage effect: negative shocks raise future volatility more than positive shocks of the same size.

Remark – Conditional vs. unconditional

Let $\mathcal{F}_{t-1} = \sigma(r_{t-1}, r_{t-2}, \dots)$ be the information set up to $t-1$.

- *Unconditional variance*: $\sigma^2 = \mathrm{Var}(r_t)$, the long-run, average risk.
- *Conditional variance*: $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$, the short-run risk knowing the past.

ARCH and GARCH models *specify* σ_t^2 while keeping σ^2 finite – the process is unconditionally homoskedastic but conditionally heteroskedastic.

Definition 1 – Mean–variance decomposition

A second-order martingale-difference innovation $\varepsilon_t = r_t - \mu_t$ is written

$$\varepsilon_t = \sigma_t\, z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} \mathcal{D}(0,1), \qquad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2.$$

$\mathcal{D}$ is the *innovation distribution*, typically $\mathcal{N}(0,1)$ or Student-t with low degrees of freedom.

1.2 ARCH(q) model

Definition 2 – ARCH(q), Engle 1982

The *autoregressive conditional heteroskedasticity* model of order q specifies

$$\varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i\, \varepsilon_{t-i}^2,$$

with $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, $\alpha_0 > 0$, and $\alpha_i \geq 0$ for $i = 1, \dots, q$.

Proposition 3 – Properties of ARCH(q)

Assume the covariance-stationarity condition $\sum_{i=1}^{q} \alpha_i < 1$. Then:

1. $\mathbb{E}[\varepsilon_t] = 0$ and (ε_t) is a white noise (uncorrelated in time, but not independent).
2. $\mathbb{E}[\varepsilon_t^2] = \alpha_0/(1 - \sum \alpha_i) =: \sigma^2$.
3. (ε_t^2) admits an AR(q) representation $\varepsilon_t^2 = \alpha_0 + \sum \alpha_i \varepsilon_{t-i}^2 + \nu_t$, with $\nu_t = \varepsilon_t^2 - \sigma_t^2$ a martingale difference.
4. Excess kurtosis: under Gaussian innovations ARCH(1) has kurtosis $3\,(1-\alpha_1^2)/(1-3\alpha_1^2) > 3$ when $\alpha_1^2 < 1/3$.

Remark – Why squares?

Squaring past residuals makes the response *sign-blind*: a -3σ surprise and a $+3\sigma$ surprise both raise σ_{t+1}^2. To allow asymmetric responses (leverage), one moves to GJR-GARCH or EGARCH (Section 1.7).

1.3 GARCH(p, q) model

In practice, ARCH(q) requires many lags q to capture the slow decay of squared-return autocorrelations, leading to long, poorly identified parameter vectors. Bollerslev (1986) generalised the model by including past conditional variances themselves.

Definition 4 – GARCH(p, q), Bollerslev 1986

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i\, \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j\, \sigma_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_i, \beta_j \geq 0$.

Proposition 5 – Stationarity of GARCH(p, q)

(ε_t) admits a covariance-stationary solution with finite unconditional variance if and only if

$$\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j \ < \ 1.$$

The unconditional variance is then

$$\sigma^2 = \mathbb{E}[\varepsilon_t^2] = \frac{\alpha_0}{1 - \sum_i \alpha_i - \sum_j \beta_j}.$$

Example 6 – GARCH$(1, 1)$: the workhorse

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

On daily equity returns, typical estimates are $\alpha_1 \approx 0.05$–0.10, $\beta_1 \approx 0.85$–0.92, $\alpha_1 + \beta_1 \approx 0.95$–$0.99$. Volatility is *very persistent* but still mean-reverting.

Remark – IGARCH and persistence

The boundary case $\alpha_1 + \beta_1 = 1$ defines an *IGARCH*: shocks have a permanent effect on the conditional variance. The unconditional variance is then infinite (the model is strictly stationary but not covariance-stationary). Estimated $\alpha_1 + \beta_1$ very close to 1 is a sign that the assumed GARCH$(1, 1)$ may be missing a structural break.

1.4 Maximum-likelihood estimation

Definition 7 – Conditional Gaussian likelihood

Assuming $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ and $r_t = \mu + \varepsilon_t$, the conditional log-likelihood is

$$\log \mathcal{L}(\boldsymbol{\theta}) = -\tfrac{T}{2} \log(2\pi) - \tfrac{1}{2} \sum_{t=1}^{T} \Big(\log \sigma_t^2(\boldsymbol{\theta}) + \frac{(r_t - \mu)^2}{\sigma_t^2(\boldsymbol{\theta})} \Big),$$

with σ_t^2 computed recursively from the GARCH equation. $\boldsymbol{\theta} = (\mu, \alpha_0, \alpha_1, \ldots, \alpha_q, \beta_1, \ldots, \beta_p)$.

Theorem 8 – Asymptotic properties of the QMLE

Under regularity conditions (positivity, stationarity, bounded fourth moment), the Gaussian quasi-MLE $\widehat{\boldsymbol{\theta}}_{\text{QMLE}}$ is consistent and asymptotically normal,

$$\sqrt{T}\,(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \ \xrightarrow{d} \ \mathcal{N}(0, \Sigma),$$

even if the true innovation distribution is non-Gaussian (e.g. Student-t). The asymptotic variance Σ takes a sandwich form $\Sigma = J^{-1} I J^{-1}$.

Remark – Practical numerics

- Returns are typically rescaled (multiplied by 100 to work in percentage units) for numerical stability.

- Initial σ_1^2 is set to the sample variance, or to the unconditional value $\alpha_0/(1-\alpha_1-\beta_1)$ at the current parameters.
- In Python, the `arch` package (`arch_model`) does the rescaling, parameter constraints and likelihood maximisation automatically.

1.5 Diagnostics

1.5.1 Engle's LM test for ARCH effects

Definition 9 – Engle's LM test

Apply on the residual $\widehat{\varepsilon}_t$ of a mean model (constant or ARMA). Run the auxiliary regression

$$\widehat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \widehat{\varepsilon}_{t-1}^2 + \cdots + \gamma_q \widehat{\varepsilon}_{t-q}^2 + u_t,$$

get its R^2. Test

$$H_0 : \gamma_1 = \cdots = \gamma_q = 0, \qquad LM = T \cdot R^2 \xrightarrow{d} \chi_q^2.$$

Reject H_0 at level α if $LM > \chi_{q,1-\alpha}^2$.

Remark – How to use it

Run Engle's LM *before* fitting GARCH (to confirm there is an ARCH effect to model) and *after* (on the standardised residuals $\widehat{z}_t = \widehat{\varepsilon}_t/\widehat{\sigma}_t$, to confirm the model has captured it).

1.5.2 Standardised residual diagnostics

Remark

A well-fitted GARCH produces standardised residuals that look like white noise both in level and in square:

- ACF of $\widehat{z}_t$ inside Bartlett bands;
- ACF of $\widehat{z}_t^2$ inside Bartlett bands;
- Engle LM on $\widehat{z}_t^2$ does not reject;
- marginal distribution close to $\mathcal{D}(0,1)$ (QQ-plot, Kolmogorov–Smirnov).

Persistent failures in $\widehat{z}_t^2$ suggest a higher-order GARCH or a Student-t innovation; failures only in $\widehat{z}_t^-$ suggest a leverage extension (GJR, EGARCH).

1.6 Volatility forecasting and Value-at-Risk

Proposition 10 – GARCH$(1,1)$ multi-step forecast

Let $\widehat{\sigma}_T^2$ and $\widehat{\varepsilon}_T$ be the values obtained at the end of the sample. Then

$$\widehat{\sigma}_{T+1}^2 = \alpha_0 + \alpha_1 \widehat{\varepsilon}_T^2 + \beta_1 \widehat{\sigma}_T^2,$$

and for $h \geq 2$,

$$\widehat{\sigma}^2_{T+h} = \sigma^2 + (\alpha_1 + \beta_1)^{h-1}(\widehat{\sigma}^2_{T+1} - \sigma^2),$$

where $\sigma^2 = \alpha_0/(1 - \alpha_1 - \beta_1)$ is the unconditional variance. The forecast *mean-reverts* to σ^2 at rate $\alpha_1 + \beta_1$.

Definition 11 – Conditional Value-at-Risk

At horizon 1 and confidence $1-\alpha$ (typically $\alpha = 1\%$ or 5%), assuming Gaussian innovations,

$$\mathrm{VaR}_{T+1}(\alpha) = -(\widehat{\mu} + z_\alpha\, \widehat{\sigma}_{T+1}),$$

with $z_\alpha = \Phi^{-1}(\alpha)$ ($z_{0.01} = -2.326$, $z_{0.05} = -1.645$). The interpretation: with probability $1 - \alpha$, the loss does not exceed $\mathrm{VaR}_{T+1}(\alpha)$.

Remark – Static vs dynamic VaR

The classical static VaR uses the *sample* standard deviation $\widehat{\sigma}$. The GARCH-VaR uses $\widehat{\sigma}_{T+1}$. In calm regimes, $\widehat{\sigma}_{T+1} < \widehat{\sigma}$ and the dynamic VaR is tighter; in turbulent regimes (just after a market shock), $\widehat{\sigma}_{T+1} > \widehat{\sigma}$ and the dynamic VaR is wider, accurately reflecting the elevated short-term risk.

1.7 Extensions (overview)

Remark – EGARCH (Nelson 1991)

The exponential GARCH models $\log \sigma_t^2$, removing the positivity constraints on the parameters and naturally allowing asymmetric responses:

$$\log \sigma_t^2 = \omega + \alpha \left(|z_{t-1}| - \mathbb{E}|z_{t-1}|\right) + \gamma\, z_{t-1} + \beta\, \log \sigma_{t-1}^2.$$

$\gamma < 0$ is the typical sign of the leverage effect on equity returns.

Remark – GJR-GARCH (Glosten–Jagannathan–Runkle 1993)

Adds a threshold term that activates only on negative shocks:

$$\sigma_t^2 = \alpha_0 + (\alpha_1 + \gamma \mathbb{1}_{\varepsilon_{t-1}<0})\varepsilon_{t-1}^2 + \beta_1\, \sigma_{t-1}^2.$$

$\gamma > 0$ implies that bad news raises future volatility more than good news of the same magnitude.

Summary

Key takeaways.

- Financial returns are uncorrelated but their squares are autocorrelated – volatility comes in clusters.
- ARCH(q): σ_t^2 is a linear function of past squared shocks. GARCH(p, q) adds memory through past conditional variances.
- GARCH(1, 1) is stationary iff $\alpha_1 + \beta_1 < 1$, with unconditional variance $\alpha_0/(1 - \alpha_1 - \beta_1)$.

- Engle's LM test detects ARCH effects in residuals; standardised residuals validate the fit.
- Conditional volatility forecasts mean-revert to the unconditional variance at rate $\alpha_1 + \beta_1$; Value-at-Risk uses these forecasts directly.
- Extensions: EGARCH and GJR-GARCH for asymmetric (leverage) effects.